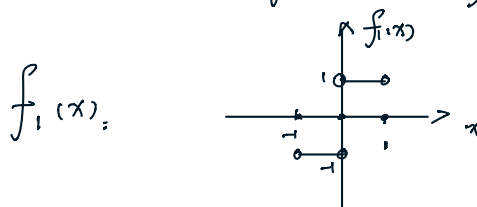


2.28. Let μ be the Lebesgue measure on $([-1,1], \mathcal{B}([-1,1]))$. For $n \geq 1$, define $f_n(x) = n \cdot \mathbb{I}_{(0, \frac{1}{n}]}(x) - n \cdot \mathbb{I}_{(-\frac{1}{n}, 0)}(x)$ and $f(x) \equiv 0$ for $x \in [-1,1]$. Show that $f_n \rightarrow f$ a.e. (μ) and $\int f_n d\mu \rightarrow \int f d\mu$ but $\{f_n\}_{n \geq 1}$ is not U.I.

To show $f_n \rightarrow f$ a.e. (μ), we need to show

$\lim_{n \rightarrow \infty} f_n(\omega) = f(\omega)$ for all $\omega \in B^c$, where $B \in \mathcal{B}([-1,1])$ s.t. $\mu(B) = 0$.

$$f_n(x) = \begin{cases} n & x \in (0, \frac{1}{n}] \\ -n & x \in (-\frac{1}{n}, 0) \\ 0 & \text{o.w.} \end{cases}$$



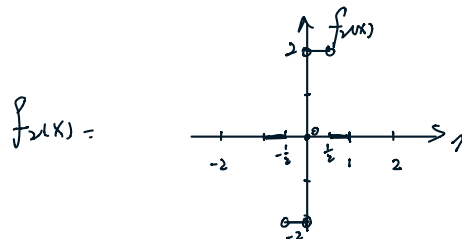
$$\forall x \in [-1,1] \setminus (-\frac{1}{n}, 0) \cup (0, \frac{1}{n}]$$

$$f_n(x) = 0 = f(x)$$

$$\text{as } n \rightarrow \infty \quad -\frac{1}{n} \rightarrow 0^-, \quad \frac{1}{n} \rightarrow 0^+$$

$$\Rightarrow (-\frac{1}{n}, 0) \cup (0, \frac{1}{n}] \rightarrow \emptyset$$

$$\Rightarrow f_n(x) \rightarrow f(x) \text{ as } n \rightarrow \infty.$$

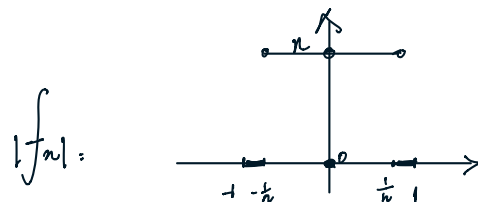


To show $\int f_n d\mu \rightarrow \int f d\mu$ but $\{f_n\}_{n \geq 1}$ is not U.I.

$$\int f_n d\mu = n \times \mu((0, \frac{1}{n}]) + (-n) \times \mu((-\frac{1}{n}, 0)) = n \times \frac{1}{n} + (-n) \times \frac{1}{n} = 1 - 1 = 0 = \int f d\mu.$$

$$\Rightarrow \int f_n d\mu \rightarrow \int f d\mu.$$

But. $\int_{\{|f_n| \geq t\}} |f_n| d\mu$



$$\forall t > 0, \exists n_0 \in \mathbb{N} \text{ s.t. } n_0 \geq t$$

$$\Rightarrow \{|f_{n_0}| \geq t\} \Leftrightarrow (-\frac{1}{n_0}, 0) \cup (0, \frac{1}{n_0}]$$

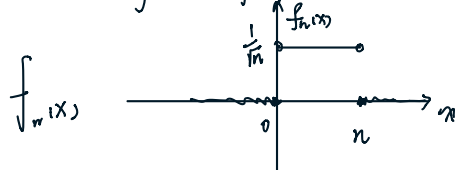
$$\Rightarrow \int_{\{|f_{n_0}| \geq t\}} |f_{n_0}| d\mu = |n_0| \times \mu((\frac{1}{n_0}, 0)) + |n_0| \times \mu((-\frac{1}{n_0}, 0)) = 1 + 1 = 2$$

$$\Rightarrow \sup_{n \geq 1} \int_{\{|f_n| \geq t\}} |f_n| d\mu = 2 \neq 0 \text{ as } t \rightarrow \infty$$

So $\{f_n\}_{n \geq 1}$ is not U.I.

2.30. For $n \geq 1$, let $f_n(x) = n^{1/2} I_{(0,n)}(x)$, $x \in \mathbb{R}$, and let $f(x) = 0$, $x \in \mathbb{R}$. Let m denote the Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Show that $f_n \rightarrow f$ a.e. (m) and $\{f_n\}_{n \geq 1}$ is UI, but $\int f_n dm \not\rightarrow \int f dm$.

To show $f_n \rightarrow f$ a.e. (m), we need $\lim_{n \rightarrow \infty} f_n(x) = f(x) \quad \forall x \in \mathbb{R}/B, \quad m(B) = 0$.



As $n \rightarrow \infty$, $\frac{1}{\sqrt{n}} \rightarrow 0 \Rightarrow f_n(x) \rightarrow 0 \quad \forall x \in \mathbb{R} \Rightarrow f_n(x) \rightarrow f(x) \quad \forall x \in \mathbb{R}$.

$\Rightarrow f_n \rightarrow f$ a.e. (m)

To show $\{f_n\}_{n \geq 1}$ is UI check $\int_{\{|f_n| \geq t\}} |f_n| dm$

$|f_n| \leq 1$ and $\forall x \in \mathbb{R} \quad f_n(x) \leq f_{n+1}(x)$

for $t \geq 1$ $|f_n| \geq t \Leftrightarrow \emptyset \Rightarrow \int_{\{|f_n| \geq t\}} |f_n| dm = 0$

$\Rightarrow \sup_{n \geq 1} \int_{\{|f_n| \geq t\}} |f_n| dm = 0$ as $t \rightarrow \infty$.

So $\{f_n\}_{n \geq 1}$ is UI.

To show $\int f_n dm \not\rightarrow \int f dm$.

$$\int f_n dm = \frac{1}{\sqrt{n}} \times \mu((0,n)) = \frac{1}{\sqrt{n}} \times n = \sqrt{n} \rightarrow \infty \neq \int f dm = 0$$

So $\int f_n dm \not\rightarrow \int f dm$.

2.36. (a) Let $\{f_n\}_{n \geq 1} \subset L^1(\mathcal{R}, \mathcal{F}, \mu)$ s.t. $f_n \rightarrow 0$ in $L^1(\mu)$ show that $\{f_n\}_{n \geq 1}$ is UI.

(b) Let $\{f_n\}_{n \geq 1} \subset L^p(\mathcal{R}, \mathcal{F}, \mu)$, $0 < p < \infty$ s.t. $\mu(\mathcal{R}) < \infty$ $\{ |f_n|^p \}_{n \geq 1}$ is UI and $f_n \xrightarrow{m} f$. show that $f \in L^p(\mu)$ and $f_n \rightarrow f$ in $L^p(\mu)$

(a) To show $\{f_n\}$ is UI, need to show $\sup_{n \geq 1} \int_{\{|f_n| \geq t\}} |f_n| d\mu \rightarrow 0$ as $t \rightarrow \infty$

$f_n \rightarrow 0$ as $n \rightarrow \infty \Rightarrow \lim_{n \rightarrow \infty} \int_{\mathcal{R}} |f_n| d\mu = 0$

$\Rightarrow \forall \varepsilon > 0, \exists n_\varepsilon \in \mathbb{N}$ s.t. $\forall n \geq n_\varepsilon \quad \int_{\mathcal{R}} |f_n| d\mu < \varepsilon$

$\Rightarrow \forall n \geq n_\varepsilon \quad \int_{\{|f_n| \geq t\}} |f_n| d\mu \leq \int_{\mathcal{R}} |f_n| d\mu < \varepsilon$

$$\Rightarrow \sup_{n \geq n_\varepsilon} \int_{\{|f_n| \geq t\}} |f_n| d\mu < \varepsilon.$$

$$\text{Also, } \{f_n\}_{n \geq 1} \subset L^1(\mathcal{R}, \mathcal{F}, \mu) \Rightarrow \forall n \geq 1 \int |f_n| d\mu < \infty.$$

$$\Rightarrow \int_{\{|f_n| \geq t\}} |f_n| d\mu \leq \int_{\mathcal{R}} |f_n| d\mu < \infty \quad \forall t > 0.$$

$$\text{by DCT } \exists t_\varepsilon > 0 \text{ s.t. } \sup_{1 \leq n \leq n_\varepsilon} \int_{\{|f_n| \geq t\}} |f_n| d\mu < \varepsilon \text{ for all } t > t_\varepsilon.$$

$$\begin{aligned} \Rightarrow \sup_{n \geq 1} \int_{\{|f_n| \geq t\}} |f_n| d\mu &= \max \left\{ \sup_{n \geq n_\varepsilon} \int_{\{|f_n| \geq t\}} |f_n| d\mu, \sup_{1 \leq n \leq n_\varepsilon} \int_{\{|f_n| \geq t\}} |f_n| d\mu \right\} \\ &\leq \max \left\{ \underbrace{\sup_{n \geq n_\varepsilon} \int_{\{|f_n| \geq t\}} |f_n| d\mu}_{< \varepsilon}, \underbrace{\sup_{1 \leq n \leq n_\varepsilon} \int_{\{|f_n| \geq t\}} |f_n| d\mu}_{< \varepsilon} \right\} \\ &< \varepsilon. \end{aligned}$$

So $\{f_n\}_{n \geq 1}$ is UI.

(b) To show $f \in L^p(\mathcal{R}, \mathcal{F}, \mu)$

$$f_n \xrightarrow{m} f \Rightarrow \exists \{f_{n_k}\}_{k \geq 1} \text{ s.t. } f_{n_k} \rightarrow f \text{ a.e. (w) as } k \rightarrow \infty.$$

$$\Rightarrow |f_{n_k}|^p \rightarrow |f|^p \text{ a.e. (w).}$$

$$\text{By } \mu(\mathcal{R}) < \infty \text{ and } \{|f_n|^p\}_{n \geq 1} \text{ is UI } \Rightarrow |f|^p \in L^1(\mu) \Rightarrow f \in L^p(\mu).$$

To show $f_n \rightarrow f$ in $L^p(\mu)$

$$\begin{aligned} \forall \varepsilon, t > 0 \quad \int |f_n - f|^p d\mu &= \int |f_n - f|^p \mathbb{1}_{\{|f_n - f|^p \leq \varepsilon\}} d\mu + \int |f_n - f|^p \mathbb{1}_{\{\varepsilon < |f_n - f|^p \leq t\}} d\mu \\ &\quad + \int |f_n - f|^p \mathbb{1}_{\{|f_n - f|^p > t\}} d\mu \\ &\leq \varepsilon^p \mu(\mathcal{R}) + t^p \mu(\{w = |f_n(w) - f(w)|^p > \varepsilon\}) + \sup_{n \geq 1} \int_{\{|f_n - f|^p > t\}} |f_n - f|^p d\mu. \end{aligned}$$

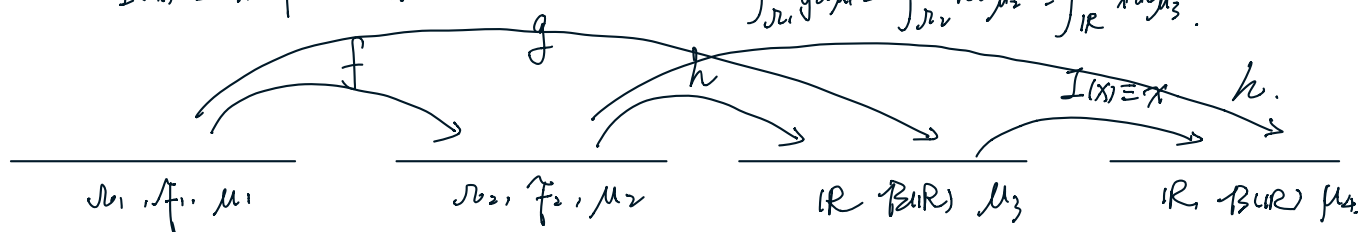
$$\Rightarrow \limsup_{n \rightarrow \infty} \int |f_n - f|^p d\mu \leq \limsup_{n \rightarrow \infty} (\varepsilon^p \mu(\mathcal{R}) + t^p \mu(\{w = |f_n(w) - f(w)|^p > \varepsilon\}) + \sup_{n \geq 1} \int_{\{|f_n - f|^p > t\}} |f_n - f|^p d\mu).$$

$$\Rightarrow \limsup_{n \rightarrow \infty} \int |f_n - f|^p d\mu \leq \varepsilon^p \mu(\mathcal{R}) + \sup_{n \geq 1} \int_{\{|f_n - f|^p > t\}} |f_n - f|^p d\mu \quad \varepsilon \rightarrow 0, \quad t \rightarrow \infty.$$

$$\Rightarrow \text{by } \{|f_n - f|^p\}_{n \geq 1} \text{ is UI and } \mu(\mathcal{R}) < \infty, \quad \limsup_{n \rightarrow \infty} \int |f_n - f|^p d\mu = 0.$$

242

Let $(\mathcal{R}_1, \mathcal{F}_1)$, $i=1,2$ be two measurable spaces. Let $f: \mathcal{R}_1 \rightarrow \mathcal{R}_2$ be $\langle \mathcal{F}_1, \mathcal{F}_2 \rangle$ -measurable, $h: \mathcal{R}_2 \rightarrow \mathbb{R}$ be $\langle \mathcal{F}_2, \mathcal{B}(\mathbb{R}) \rangle$ -measurable, and μ_1 be a measure on $(\mathcal{R}_1, \mathcal{F}_1)$. Show that $g \equiv h \circ f$, i.e. $g(\omega) = h(f(\omega))$ for $\omega \in \mathcal{R}_1$ is in $L^1(\mu_1)$ iff $h(\cdot) \in L^1(\mathcal{R}_2, \mathcal{F}_2, \mu_2)$ where $\mu_2 = \mu_1 f^{-1}$ iff $I(\cdot) \in L^1(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu_3)$, $\mu_3 \equiv \mu_2 h^{-1}$ where $I(\cdot)$ is the identity function in $\mathbb{R}$, i.e. $I(x) \equiv x$ for all $x \in \mathbb{R}$ and also that $\int_{\mathcal{R}_1} g d\mu_1 = \int_{\mathcal{R}_2} h d\mu_2 = \int_{\mathbb{R}} x d\mu_3$.



$$\mu_2 = \mu_1(f^{-1}) \quad , \quad \mu_3 = \mu_2(h^{-1}) \quad , \quad \mu_4 = \mu_3(I^{-1}) = \mu_3 \quad , \quad g = h(f)$$

To show $g \in L^1(\mu_1) \Leftrightarrow h \in L^1(\mu_2)$ and $\int_{\mathcal{R}_1} g d\mu_1 = \int_{\mathcal{R}_2} h d\mu_2$

First, suppose h to be any indicator function I_A , $A \in \mathcal{F}_2$.

$$I_A \in L^1(\mu_2) \Leftrightarrow \int_{\mathcal{R}_2} I_A(x) d\mu_2(x) < +\infty \Leftrightarrow \mu_2(A) = \mu_1(f^{-1}(A)) < +\infty.$$

$$\Leftrightarrow \int_{\mathcal{R}_1} I_{f^{-1}(A)}(\omega) d\mu_1(\omega) < +\infty \Leftrightarrow I_A(f(\cdot)) \in L^1(\mu_1)$$

$$\text{Also, } \int_{\mathcal{R}_1} I_A(f(\omega)) d\mu_1(\omega) = \int_{\mathcal{R}_1} I_{f^{-1}(A)}(\omega) d\mu_1(\omega) = \mu_1(f^{-1}(A)) = \mu_2(A) = \int_{\mathcal{R}_2} I_A(x) d\mu_2(x).$$

$$\Rightarrow g \in L^1(\mu_1) \Leftrightarrow h \in L^1(\mu_2) \quad \text{and} \quad \int_{\mathcal{R}_1} g d\mu_1 = \int_{\mathcal{R}_2} h d\mu_2 \quad \text{is true for } h \text{ being } I_A$$

$\Rightarrow$ this is true for h being any simple nonnegative measurable functions.

Secondly, Suppose that $h(\cdot)$ is any nonnegative measurable function,

$$\Rightarrow \exists \{h_n\}_{n=1}^\infty \text{ s.t. } \forall \omega \in \mathcal{R}_2 \quad h_n(\omega) \leq h_{n+1}(\omega) \quad \text{and} \quad \lim_{n \rightarrow \infty} h_n(\omega) = h(\omega)$$

$$\text{By MCT } \int_{\mathcal{R}_2} h d\mu_2 = \lim_{n \rightarrow \infty} \int_{\mathcal{R}_2} h_n d\mu_2 = \lim_{n \rightarrow \infty} \int_{\mathcal{R}_1} h_n \circ f d\mu_1 = \int_{\mathcal{R}_1} h \circ f d\mu_1 = \int_{\mathcal{R}_1} g d\mu_1$$

$$\Rightarrow g \in L^1(\mu_1) \Leftrightarrow h \in L^1(\mu_2) \quad \text{and} \quad \int_{\mathcal{R}_1} g d\mu_1 = \int_{\mathcal{R}_2} h d\mu_2 \quad \text{is true for } h \text{ being any non-negative measurable functions.}$$

Thirdly, Suppose $h(\cdot)$ is any measurable function

$$h \in L^1(\mathcal{R}_2) \Leftrightarrow h^+, h^- \in L^1(\mathcal{R}_2) \Leftrightarrow h^+ \circ f, h^- \circ f \in L^1(\mathcal{R}_1) \Leftrightarrow h \circ f = g \in L^1(\mathcal{R}_1)$$

$$\begin{aligned} \text{when } h \in L^1(\mathcal{R}_2) \Rightarrow \int_{\mathcal{R}_2} h d\mu_2 &= \int_{\mathcal{R}_2} h^+ d\mu_2 - \int_{\mathcal{R}_2} h^- d\mu_2 = \int_{\mathcal{R}_1} h^+ \circ f d\mu_1 - \int_{\mathcal{R}_1} h^- \circ f d\mu_1 \\ &= \int_{\mathcal{R}_1} (h^+ - h^-) \circ f d\mu_1 = \int_{\mathcal{R}_1} g d\mu_1 \end{aligned}$$

To show $\int_{\mathbb{R}_2} h d\mu_2 = \int_{\mathbb{R}} x d\mu_3$. and $h \in L^1(\mu_2) \Leftrightarrow I \in L^1(\mu_3)$

set $I(x)$ as $h(\cdot)$, h as $f(\cdot)$, h as $I \circ h = h(f(\cdot)) = g(\cdot)$.

μ_2 as μ_1 μ_3 as μ_2

directly we know $h \in L^1(\mu_2) \Leftrightarrow I \in L^1(\mu_3)$

$$\text{and } \int_{\mathbb{R}_2} h d\mu_2 = \int_{\mathbb{R}} I d\mu_3 = \int_{\mathbb{R}} x d\mu_3.$$